Challenge Owner

PETROVIETNAM

Ambient Systems for PetroVietnam

Integrated technical agency, driving financial performance for world's most competitive brands.

Entry:

AMBIENT SYSTEMS

Co-organizers

METI
Ministry of Economy, Trade and Industry

JETRO
Japan External Trade Organization

Company Overview

Ambient Systems: Global & Borderless

Integrated Deep-Tech Expertise -- Engineering, FP&A, AI & IP | From Asia-Pacific Hub to

A Lean, Borderless Agency Delivering High-Impact Results Globally

- ✓ Born from 5 years of award-winning NUS deep-tech R&D (PCT-patent, platform prototype @ TRL 6)
- ✓ Now operating from HK hub with Belgrade R&D and international networks (Japan, Europe).

3+

Globally-competitive Regions (Asia, Americas, and Europe)

10+

Combined Experience (Founder + Company)

2

Corporations

$150k

Invested in R&D

2020

Year Founded

1

Delivery node (via HK)

Ambient Systems: Borderless consulting that turns deep-tech innovation into measurable financial & operational gains – anywhere, anytime.

Introduction

Agency Overview

Technical agency driving financial performance through engineering, artificial intelligence, and intellectual property.

Offers:

- ✓ AI applications for industries
- ✓ Streamlined financial planning & analysis (FP&A)
- ✓ Intellectual Property research & development (R&D)
- ✓ Software subscription (in development)

Services:

1. **Civil Engineering & Energy Systems**
   Drafting and load calculations.
   Solar array design, microgrid optimization, building energy efficiency.

2. **Supply-Chain Management & Vendor Optimization**
   Sourcing strategy, customs optimization, vendor scorecards.

3. **Vertical Farming / CEA Optimization**
   Layout design, yield modeling, environmental controls.

4. **IP Strategy & Patent Support**
   Prior art searches, patent filing support, freedom-to-operate analysis.

Case Study

End-to-End Climate Management Solution for Indoor Farming

BUSINESS CHALLENGE

Indoor farmers face significant crop loss and high operational costs due to the inability to maintain optimal climate conditions. Existing solutions are fragmented, manual, and labor-intensive, often leading to unmonitored temperature and humidity imbalances that promote mold and pests, ultimately reducing yield quality and quantity.

APPROACH TO SOLUTION

- Defined the full-stack ML solution (sensors, optimization engine, dashboard) and the scalable B2B SaaS + Hardware business model.
- Secured pilot Letters of Intent (LOIs) with two key vertical farm operators to validate demand and establish the initial pipeline.
- Developed the financial model projecting a 24% improvement in client margins with a rapid 4–6-month customer payback period.
- Produced all investment deliverables (Investment Deck, Business Plan) and defined the IP/patent strategy for the core optimization algorithm.

OUTCOMES ACHIEVED

- 24% ROI Model: Developed a model projecting a 24% improvement in client margins with a rapid 4–6-month customer payback.
- Validation: Achieved critical market validation and established a pipeline through secured commercial LOIs.
- IP Foundation: Defined the patent strategy for the core optimization algorithm.

AMBIENT SYSTEMS PoC/Solution

PETROVIETNAM

Intelligent Twin (PoC)

Solution for PetroVietnam (PVN)

Current Status

- Company: PetroVietnam
- Location: Offshore platform #421 (TBC)
- Department: Operations/Finance (TBC)
- Challenge #1: Complex business processes involving multiple approvals across departments result in prolonged processing times and obscure bottlenecks
- Collaboration Need #3: An intelligent digital twin

Proposal

Week 1-2: Extraction

- Ingest Anonymized Samples (Dossiers)
- Define Schema
- Deliverable: Knowledge Status Quo report.

Week 2-3: Insertion

- Create a digital twin knowledge graph using Cypher Patterns to map inter-departmental dependencies.
- Mirror the selected process (e.g., Sales or Procurement) in Neo4j.
- Deliverable: Functional "Process" Knowledge Graph.

Week 3-4: The AI Layer

- Run AI agent scenarios (The "Policy Sandbox") to test regulatory changes via GraphRAG.
- Identify Parallel Processing paths to resolve the 20-day lags found in the graph.
- Deliverable: Risk-Free Policy Simulation Report.

Outcomes

- Identified Bottlenecks: A data-driven heat map pinpointing structural "lags" discovered through Cypher Pattern analysis of inter-departmental dependencies.
- Sandbox Simulation Results: Validated proof of 15%+ efficiency gains by testing "What-If" scenarios, such as shifting from sequential to parallel processing paths.
- Unified Governance Dashboard: A functional "Digital Twin" interface (Ambient Platform) providing real-time visibility into "Data Status" and automated AI Recommendations for workflow re-routing.

PVN: Organizational Digital Twin for Intelligent Governance

Proposed Solution: Multi-Agent AI & Knowledge Graph Platform







	Step 1	Step 2	Step 3
Step	Process manuals & anonymized dossiers	Build Knowledge Graph of dept. dependencies	AI for "What-If" testing.
Details	Create a "Living Foundation" by mapping Operations → Finance.	Use Cypher Patterns to detect lags.	Identify Parallel Processing paths to cut cycle time.
Tech	• Python, Process Mapping	• Neo4j,	, Gemini/OpenAI

Measurable Results: reduce bottlenecks by 15%+.

Direct Access to Award-Winning Technical Leadership

Your Lead Architect for the PVN Governance Twin

Industrial Record	Award-Winning Tech	Global Delivery Hub
• 10+ Years Asia-Pacific • Scaled YC-backed ventures. • 12% OpEx Reduction in heavy industry/mining.	• Panasonic Deep Tech Winner • PCT-Awarded PoC • Google -up	• HK Set-up • HK Top Talent Pass for on-site • America's regional presence for potential partnerships

Chief Systems Architect & Founder

PhD Researcher
(Business Management)

M.Sc. Computer Science

Intellectual Property Expert.

Integrated Regional Delivery: Utilizing the Hong Kong hub for immediate on-site deployment in Vietnam and Japan (Authorized via HK Top Talent Pass).

Why Ambient Systems for Inno Vietnam–Japan

Ready for Japan-Vietnam Co-Dev via JETRO Networks

Proven impact in global supply-chain and energy projects

- 12 % OpEx reduction (client)
- 24 % ROI (Singapore-based vertical farming project)

Japanese Collaborations

- City.Tech Tokyo Demonstration (Singapore delegation)
- Panasonic Deep Tech Innovation Challenge Winner (Singapore)

Globally validated tech

- Google (Singapore) Runner-Up
- WIPO PCT
- NSERC-Funded R&D (Quebec, Canada)

Invoiced globally from Hong Kong entity (set-up in progress)

- Delivered remotely
- Singapore mailing address available
- Immediately available for on-site or remote collaboration

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